10.2. SIMPLE MACHINE

In a broad sense, a simple machine may be defined as a device, which enables us to do some useful work at some point or to overcome some resistance, when an effort or force is applied to it, at some other convenient point.

10.3. COMPOUND MACHINE

A compound machine may be defined as a device, consisting of a number of simple machines which enables us to do some useful work at a faster speed or with a much less effort as compared to a simple machine.

10.4. LIFTING MACHINE

It is a device, which enables us to lift a heavy load (W) by applying a comparatively smaller effort (P).

10.5. MECHANICAL ADVANTAGE

The mechanical advantage (briefly written as M.A.) is the ratio of weight lifted (W) to the effort applied (P) and is always expressed in pure number. Mathematically, mechanical advantage,

$$\text{M.A.} = \frac{W}{P}$$

10.6. INPUT OF A MACHINE

The input of a machine is the work done on the machine. In a lifting machine, it is measured by the product of effort and the distance through which it has moved.

10.7. OUTPUT OF A MACHINE

The output of a machine is the actual work done by the machine. In a lifting machine, it is measured by the product of the weight lifted and the distance through which it has been lifted.

10.8. EFFICIENCY OF A MACHINE

It is the ratio of output to the input of a machine and is generally expressed as a percentage. Mathematically, efficiency,

$$\eta = \frac{\text{Output}}{\text{Input}} \times 100$$

10.9. IDEAL MACHINE

If the efficiency of a machine is 100% *i.e.*, if the output is equal to the input, the machine is called as a perfect or an *ideal machine.*

10.10. VELOCITY RATIO

The velocity ratio (briefly written as V.R.) is the ratio of distance moved by the effort (y) to the distance moved by the load (x) and is always expressed in pure number. Mathematically, velocity ratio,

$$\text{V.R.} = \frac{y}{x}$$

10.11. RELATION BETWEEN EFFICIENCY, MECHANICAL ADVANTAGE AND VELOCITY RATIO OF A LIFTING MACHINE

It is an important relation of a lifting machine, which throws light on its mechanism. Now consider a lifting machine, whose efficiency is required to be found out.

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Let W = Load lifted by the machine,
P = Effort required to lift the load,
Y = Distance moved by the effort, in lifting the load, and
x = Distance moved by the load.

We know that $\text{M.A.} = \frac{W}{P} = W/P$ and $\text{V.R.} = \frac{y}{x} = y/x$

We also know that input of a machine
= Effort applied × Distance through which the effort has moved
$= P \times y$ ...(*i*)

and output of a machine = Load lifted × Distance through which the load has been lifted
$= W \times x$ ...(*ii*)

∴ Efficiency, $\eta = \frac{\text{Output}}{\text{Input}} = \frac{W \times x}{P \times y} = \frac{W/P}{y/x} = \frac{\text{M.A.}}{\text{V.R.}}$

Note. It may be seen from the above relation that the values of M.A. and V.R. are equal only in case of a machine whose efficiency is 100%. But in actual practice, it is not possible.

Example 10.1. *In a certain weight lifting machine, a weight of 1 kN is lifted by an effort of 25 N. While the weight moves up by 100 mm, the point of application of effort moves by 8 m. Find mechanical advantage, velocity ratio and efficiency of the machine.*

Solution. Given: Weight (W) = 1 kN = 1000 N ; Effort (P) = 25 N ; Distance through which the weight is moved (x) = 100 mm = 0.1 m and distance through which effort is moved (y) = 8 m.

Mechanical advantage of the machine.

We know that mechanical advantage of the machine

$$\text{M.A.} = \frac{W}{P} = \frac{1000}{25} = 40 \quad \textbf{Ans.}$$

Velocity ratio of the machine

We know that velocity ratio of the machine

$$\text{V.R.} = \frac{y}{x} = \frac{8}{0.1} = 80 \quad \textbf{Ans.}$$

Efficiency of the machine

We also know that efficiency of the machine,

$$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{40}{80} = 0.5 = 50\% \quad \textbf{Ans.}$$

10.12. REVERSIBILITY OF A MACHINE

Sometimes, a machine is also capable of doing some work in the reversed direction, after the effort is removed. Such a machine is called a reversible machine and its action is known as *reversibility of the machine.*

10.13. CONDITION FOR THE REVERSIBILITY OF A MACHINE

Consider a reversible machine, whose condition for the reversibility is required to be found out.

Let W = Load lifted by the machine,
P = Effort required to lift the load,
y = Distance moved by the effort, and
x = Distance moved by the load.

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We know that input of the machine

$$= P \times y \quad \ldots(i)$$

and output of the machine $= W \times x \quad \ldots(ii)$

We also know that machine friction

$$= \text{Input} - \text{Output} = (P \times y) - (W \times x) \quad \ldots(iii)$$

A little consideration will show that in a reversible machine, the *output of the machine should be more than the machine friction, when the effort (P) is zero. *i.e.*,

$$W \times x > P \times y - W \times x$$

or $$2\,W \times x > P \times y$$

or $$\frac{W \times x}{P \times y} > \frac{1}{2}$$

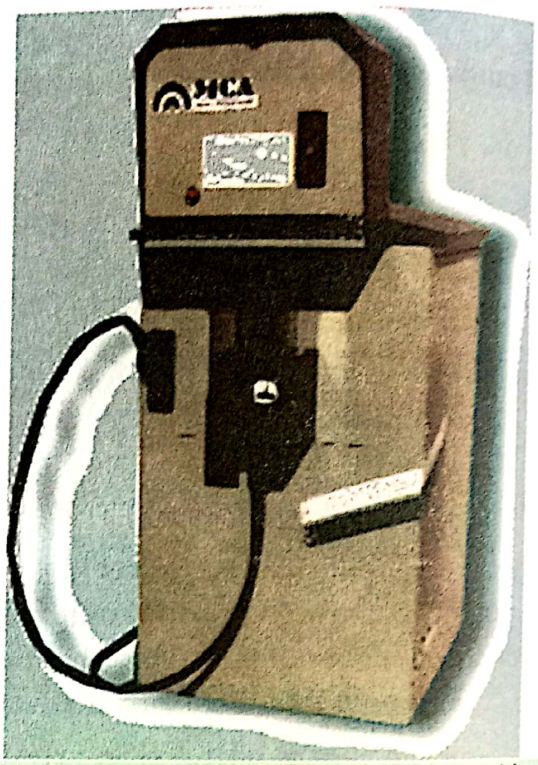
or $$\frac{\frac{W}{P}}{\frac{y}{x}} > \frac{1}{2}$$

or $$\frac{\text{M.A.}}{\text{V.R.}} > \frac{1}{2}$$

$$\ldots\left(\because \frac{W}{P} = \text{M.A.} \quad \text{and} \quad \frac{y}{x} = \text{V.R.}\right)$$

$\therefore$ $$\eta > \frac{1}{2} = 0.5 = 50\%$$

Hence the condition for a machine, to be reversible, is that its efficiency should be *more than* 50%.



Oil extractor for petrol station, car washing area and garages is irreversible machine.

10.14. SELF-LOCKING MACHINE

Sometimes, a machine is not capable of doing any work in the reversed direction, after the effort is removed. Such a machine is called a non-reversible or *self-locking machine*. A little consideration will show, that the condition for a machine to be non-reversible or self-locking is that its efficiency should *not be more than* 50%.

Example 10.2. *A certain weight lifting machine of velocity ratio 30 can lift a load of 1500 N with the help of 125 N effort. Determine if the machine is reversible.*

Solution. Given: Velocity ratio (V.R.) = 30; Load (W) = 1500 N and effort (P) = 125 N.

We know that $$\text{M.A.} = \frac{W}{P} = \frac{1500}{125} = 12$$

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{12}{30} = 0.4 = 40\%$$

Since efficiency of the machine is less than 50%, therefore the machine is non-reversible. Ans.

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10.17. MAXIMUM MECHANICAL ADVANTAGE OF A LIFTING MACHINE

We know that mechanical advantage of a lifting machine,

$$\text{M.A.} = \frac{W}{P}$$

For maximum mechanical advantage, substituting the value of $P = mW + C$ in the above equation,

$$\text{Max. M.A.} = \frac{W}{mW + C} = \frac{1}{m + \frac{C}{W}} = \frac{1}{m} \quad \ldots \left(\text{Neglecting} \right.$$

10.18. MAXIMUM EFFICIENCY OF A LIFTING MACHINE

We know that efficiency of a lifting machine,

$$\eta = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}} = \frac{\frac{W}{P}}{\text{V.R.}} = \frac{W}{P \times \text{V.R.}}$$

For *maximum efficiency, substituting the value of $P = mW + C$ in the above equation,

$$\text{Max. } \eta = \frac{W}{(mW + C) \times \text{V.R}} = \frac{1}{\left(m + \frac{C}{W} \right) \times \text{V.R.}} = \frac{1}{m \times \text{V.R.}} \quad \ldots \left(\text{Neglecting} \right.$$

Example 10.8. *The law of a machine is given by the relation :*

$$P = 0.04\ W + 7.5$$

where (P) is the effort required to lift a load (W), both expressed in newtons. What is the mechan advantage and efficiency of the machine, when the load is 2 kN and velocity ratio is 40 ? What is maximum efficiency of the machine ?

If (F) is the effort lost in friction, find the relation between F and W. Also find the value o when W is 2 kN.

Solution. Given: Law of machine $P = 0.04\ W + 7.5$; Load $(W) = 2$ kN $= 2000$ N and veloc ratio (V.R.) $= 40$.

---

* We know that efficiency of a lifting machine,

$$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$

A little consideration will show that the efficiency will be maximum, when the mechanical advanta will be maximum.

or

$$\text{Max. } \eta = \frac{\text{Max. M.A.}}{\text{V.R.}}$$

$$= \frac{1}{m \times \text{V.R.}} \quad \ldots \left(\because \text{Max. M.A.} = \frac{1}{m} \right.$$

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Mechanical advantage

Substituting the value of W in the law of the machine,

$$P = mW + C = 0.04 \times 2000 + 7.5 = 87.5 \text{ N}$$

$$\therefore \quad \text{M.A.} = \frac{W}{P} = \frac{2000}{87.5} = 22.9 \qquad \textbf{Ans.}$$

Efficiency of the machine

We know that $\eta = \dfrac{\text{M.A.}}{\text{V.R.}} = \dfrac{22.9}{40} = 0.5725 = 57.25\%$ **Ans.**

Maximum efficiency of the machine

We know that *maximum efficiency of the machine,

$$\text{Max.}\eta = \frac{1}{m \times \text{V.R.}} = \frac{1}{0.04 \times 40} = 0.625 = 62.5\% \qquad \textbf{Ans.}$$

Relation between F and W

We know that effort lost in friction,

$$F_{(\text{effort})} = P - \frac{W}{\text{V.R.}} = (0.04\,W + 7.5) - \frac{W}{\text{V.R.}}$$

$$= W\left(0.04 - \frac{1}{\text{V.R.}}\right) + 7.5 = W\left(0.04 - \frac{1}{40}\right) + 7.5$$

$$= W\,(0.04 - 0.025) + 7.5 = 0.015\,W + 7.5 \qquad \textbf{Ans.}$$

Value of F when W is 2 kN

Substituting the value of W equal to 2 kN or 2000 N in the above equation,

$$F = (0.015 \times 2000) + 7.5 = 37.5 \text{ N} \qquad \textbf{Ans.}$$

Example 10.9. *The law of a certain lifting machine is :*

$$P = \frac{W}{50} + 8$$

The velocity ratio of the machine is 100. Find the maximum possible mechanical advantage and the maximum possible efficiency of the machine. Determine the effort required to overcome the machine friction, while lifting a load of 600 N. Also calculate the efficiency of the machine at this load.

Solution. Given: Law of lifting machine $P = \dfrac{W}{50} + 8 = 0.02W + 8$; Velocity ratio (V.R.) = 100 and load (W) = 600 N.

Maximum possible mechanical advantage

Comparing the given law of the machine with the standard relation for the law of the machine (*i.e.* $P = mW + C$) we find that in the given law of the machine, $m = 0.02$. We know that maximum possible mechanical advantage

$$\text{Max M.A.} = \frac{1}{m} = \frac{1}{0.02} = 50 \qquad \textbf{Ans.}$$

Maximum possible efficiency

We know that maximum possible efficiency

$$= \frac{1}{m \times \text{V.R.}} = \frac{1}{0.02 \times 100} = \frac{1}{2} = 0.5 = 50\% \qquad \textbf{Ans.}$$

* In this example, the value of $m = 0.04$ and $C = 7.5$ as the law of the machine is given by the relation $P = mW + C$.

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12. Worm geared

11.3. SIMPLE WHEEL AND AXLE

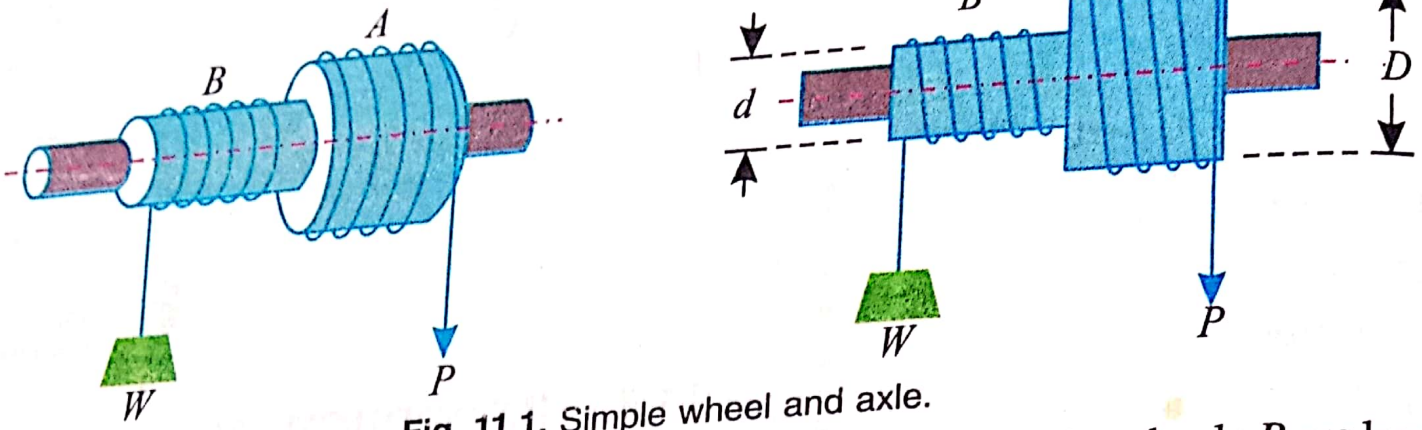


Fig. 11.1. Simple wheel and axle.

In Fig. 11.1 is shown a simple wheel and axle, in which the wheel A and axle B are keyed to the same shaft. The shaft is mounted on ball bearings, order to reduce the frictional resistance to a minimum. A string is wound round the axle B, which carries the load to be lifted. A second string is wound round the wheel A in the *opposite direction* to that of the string on B.

Let D = Diameter of effort wheel,

d = Diameter of the load axle,

W = Load lifted, and

P = Effort applied to lift the load.

One end of the string is fixed to the wheel, while the other is free and the effort is applied to this end. Since the two strings are wound in opposite directions, therefore a downward motion of the effort (P) will raise the load (W).

Since the wheel as well as the axle are keyed to the same shaft, therefore when the wheel rotates through one revolution, the axle will also rotate through one revolution. We know that displacement of the effort in one revolution of effort wheel A,

$$= \pi D \qquad \ldots(i)$$

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and displacement of the load in one revolution $= \pi d$

$$\therefore \quad \text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}} = \frac{\pi D}{\pi d} = \frac{D}{d} \qquad ...(ii)$$

Now $$\text{M.A.} = \frac{\text{Load lifted}}{\text{Effort applied}} = \frac{W}{P}$$

and efficiency $$\eta = \frac{\text{M.A}}{\text{V.R.}} \qquad ...\text{as usual}$$

Example 11.1. *A simple wheel and axle has wheel and axle of diameters of 300 mm and 30 mm respectively. What is the efficiency of the machine, if it can lift a load of 900 N by an effort of 100 N.*

Solution. Given: Diameter of wheel (D) = 300 mm; Diameter of axle (d) = 30 mm; Load lifted by the machine (W) = 900 N and effort applied to lift the load (P) = 100 N

We know that velocity ratio of the simple wheel and axle,

$$\text{V.R.} = \frac{D}{d} = \frac{300}{30} = 10$$

and mechanical advantage $$\text{M.A.} = \frac{W}{P} = \frac{900}{100} = 9$$

$$\therefore \quad \text{Efficiency, } \eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{9}{10} = 0.9 = 90\% \quad \textbf{Ans.}$$

Example 11.2. *A drum weighing 60 N and holding 420N of water is to be raised from a well by means of wheel and axle. The axle is 100 mm diameter and the wheel is 500 mm diameter. If a force of 120 N has to be applied to the wheel, find (i) mechanical advantage, (ii) velocity ratio and (iii) efficiency of the machine.*

Solution. Given: Total load to be lifted (W) = 60 + 420 = 480 N; Diameter of the load axle (d) = 100 mm; Diameter of effort wheel (D) = 500 mm and effort (P) = 120 N.

Mechanical advantage

We know that mechanical advantage

$$\therefore \quad \text{M.A.} = \frac{W}{P} = \frac{480}{120} = 4 \quad \textbf{Ans.}$$

Velocity ratio

We know that velocity ratio

$$\text{V.R.} = \frac{D}{d} = \frac{500}{100} = 5 \quad \textbf{Ans.} \qquad ...(ii)$$

Efficiency of the machine

We also know that efficiency of the machine,

$$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{4}{5} = 0.8 = 80\% \quad \textbf{Ans.}$$

Note : If we consider weight of the water only (*i.e.*, neglecting weight of the drum) then

$$\text{M.A.} = \frac{420}{120} = 3.5 \quad \textbf{Ans.}$$

and efficiency $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{3.5}{5} = 0.7 = 70\% \quad \textbf{Ans.}$$

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11.4. DIFFERENTIAL WHEEL AND AXLE

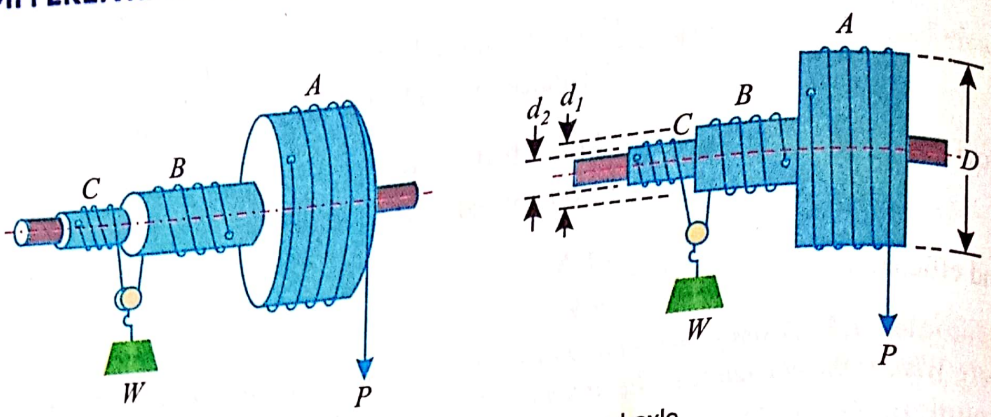


Fig. 11.2. Differential wheel and axle.

It is an improved form of simple wheel and axle, in which the velocity ratio is intensified with the help of load axle. In fig. 11.2 is shown a differential wheel and axle. In this case, the load axle BC is made up of two parts of different diameters. Like simple wheel and axle, the wheel A, and the axles B and C are keyed to the same shaft, which is mounted on ball bearings in order to reduce the frictional resistance to a minimum.

The effort string is wound round the wheel A. Another string wound round the axle B, which after passing round the pulley (to which the weigt W is attached) is wound round the axle C in opposite direction to that of the axle B; *care being taken to wind the string on the wheel A and axle C in the same direction.* As a result of this, when the string unwinds from the wheel A, the other string also unwinds from the axle C. But it winds on the axle B as shown in Fig. 11.2.

Let D = Diameter of the effort wheel A,

d_1 = Diameter of the axle B,

d_2 = Diameter fo the axle C,

W = Weight lifted by the machine, and

P = Effort applied to lift the weight.

We know that displacement of the effort in one revolution of effort wheel A

$$= \pi D \qquad \text{...(i)}$$

$\therefore$ Length of string, which will wound on axle B in one revolution

$$= \pi d_1$$

and length of string, which will unwound from axle C in one revolution

$$= \pi d_2$$

$\therefore$ Length of string which will wound in one revolution

$$= \pi\, d_1 - \pi\, d_2 = \pi\,(d_1 - d_2)$$

and displacement of weight $$= \frac{1}{2} \times \pi(d_1 - d_2) = \frac{\pi}{2}(d_1 - d_2) \qquad \text{...(ii)}$$

$\therefore$ $$\text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}} = \frac{\pi D}{\frac{\pi}{2}(d_1 - d_2)} = \frac{2D}{d_1 - d_2}$$

Now $$\text{M.A.} = \frac{W}{P}$$ ...as usual

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$ ...as usual

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Example 11.3. *The larger and smaller diameters of a differential wheel and axle are 80 mm and 70 mm respectively. The effort is applied to the wheel of diameter 250 mm. What is the velocity ratio? Find efficiency and frictional effort lost, when a load of 1050 N is lifted by an effort of 25 N.*

Solution. Given: Larger diameter of wheel (d_1) = 80 mm; Smaller diameter of wheel (d_2) = 70 mm; Diameter of the effort wheel (D) = 250 mm; Load lifted (W) = 1050 N and effort (P) = 25 N.

Velocity ratio

We know that velocity ratio

$$\text{V.R.} = \frac{2D}{d_1 - d_2} = \frac{2 \times 250}{80 - 70} = 48 \quad \textbf{Ans.} \qquad ...(i)$$

Efficiency

We know that mechanical advantage

$$\text{M.A.} = \frac{W}{P} = \frac{1050}{25} = 42 \qquad ...(ii)$$

and efficiency,

$$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{42}{50} = 0.84 = 84\% \quad \textbf{Ans.}$$

Frictional effort lost

We also know that frictional effort lost,

$$F_{(\text{effort})} = P - \frac{W}{\text{V.R.}} = 25 - \frac{1050}{50} = 4 \text{ N} \quad \textbf{Ans.}$$

Example 11.4. *With a differential wheel and axle, an effort of 6 N raised a load of 60 N. If the efficiency at this load is 80%, find the velocity ratio of the machine.*

If the diameter of the effort wheel is 300 mm, determine the difference betwen the diameters of the axles. If the sum of the diameters of the axles is 280 mm, determine the diameter of each axle.

Solution: Given: Effort (P) = 6 N; Load raised (W) = 60 N; Efficiency (η) = 80% = 0.8; Diameter of effort wheel (D) = 300 mm and sum of the diameters of axles ($d_1 + d_2$) = 280 mm.

Velocity ratio of the machine

We know that mechanical advantage

$$\text{M.A.} = \frac{W}{P} = \frac{60}{6} = 10$$

and efficiency,

$$0.8 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{10}{\text{V.R.}}$$

$$\therefore \quad \text{V.R.} = \frac{10}{0.8} = 12.5 \quad \textbf{Ans.}$$

Difference between the diameters of the axles

We know that velocity ratio of a differential wheel and axle,

$$12.5 = \frac{2D}{d_1 - d_2} = \frac{2 \times 300}{d_1 - d_2} = \frac{600}{d_1 - d_2}$$

$$\therefore \quad (d_1 - d_2) = \frac{600}{12.5} = 48 \quad \textbf{Ans.}$$

Diameter of each axle

Solving $(d_1 - d_2) = 48$ and $(d_1 + d_2) = 280$ simultaneously, we get

$$d_1 = 164 \text{ mm and } d_2 = 116 \text{ mm} \quad \textbf{Ans.}$$

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Example 11.5. *In a differential wheel and axle, the diameter of the effort wheel is 400 mm. The radii of the axles are 150 mm and 100 mm respectively. The diameter of the rope is 10 mm. Find the load which can be lifted by an effort of 25 N assuming the efficiency of the machine to be 84%.*

Solution. Given: Diameter of effort wheel = 480 mm; Radii of axles = 150 mm and 100 mm or diameter of axles = 300 mm and 200 mm; Diameter of rope = 10 mm; Effort (P) = 25 N and efficiency (η) = 84% = 0.84.

Let W = Load that can be lifted by the machine.

We know that effective diameter of the effort wheel,

$$D = 400 + 10 = 410 \text{ mm}$$

and effective diameters of axles,

$$d_1 = 300 + 10 = 310 \text{ mm and } d_2 = 200 + 10 = 210 \text{ mm}$$

We also know that velocity ratio of a differential wheel and axle,

$$\text{V.R.} = \frac{2D}{d_1 - d_2} = \frac{2 \times 410}{310 - 210} = 8.2$$

and

$$\text{M.A.} = \frac{W}{P} = \frac{W}{25}$$

We also know that efficiency,

$$84 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{\frac{W}{25}}{8.2} = \frac{W}{205}$$

or $$W = 0.84 \times 205 = 172.2 \text{ N}$$ **Ans.**

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or

11.7. WORM AND WORM WHEEL

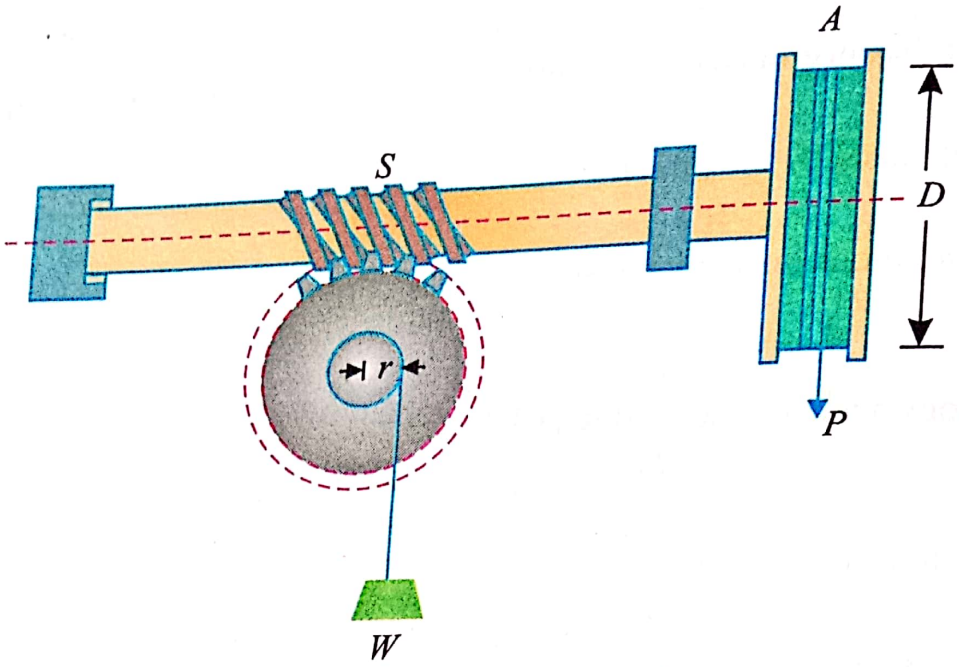


Fig. 11.5. Worm and worm wheel

It consists of a square threaded screw, S (known as worm) and a toothed wheel (known as worm wheel) geared with each other, as shown in Fig. 11.5. A wheel A is attached to the worm, over which passes a rope as shown in the figure. Sometimes, a handle is also fixed to the worm (instead of the wheel). A load drum is securely mounted on the worm wheel.

Let D = Diameter of the effort wheel,

r = Radius of the load drum

W = Load lifted,

P = Effort applied to lift the load, and

T = No. of teeth on the worm wheel.

We know that distance moved by the effort in one revolution of the wheel (or handle)

$$= \pi D \qquad ...(i)$$

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If the worm is single-threaded (*i.e.*, for one revolution of the wheel *A*, the screw *S* pushes the worm wheel through one teeth), then the load drum will move through

$$= \frac{1}{T} \text{ revolution}$$

and distance, through which the load will move

$$= \frac{2\pi r}{T} \qquad ...(ii)$$

$$\therefore \quad \text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}}$$

$$= \frac{\pi D}{\frac{2\pi r}{T}} = \frac{DT}{2r} \qquad ...(iii)$$

Now $$\text{M.A.} = \frac{W}{P} \qquad ...\text{as usual}$$

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} \qquad ...\text{as usual}$$

Notes : 1. If the worm is double-threaded *i.e.*, for one revolution of wheel *A*, the screw *S* pushes the worm wheel through two teeths, then

$$\text{V.R.} = \frac{DT}{2 \times 2r} = \frac{DT}{4r}$$

2. In general, if the worm is *n* threaded, then

$$\text{V.R.} = \frac{DT}{2nr}$$

Example 11.9. *A worm and worm wheel with 40 teeth on the worm wheel has effort wheel of 300 mm diameter and load drum of 100 mm diameter. Find the efficiency of the machine, if it can lift a load of 1800 N with an effort of 24 N.*

Solution. Given: No. of teeth on the worm wheel (T) = 40 ; Diameter of effort wheel = 300 mm Diameter of load drum = 100 mm or radius (r) = 50 mm; Load lifted (W) 1800 N and effort (P) = 24 N.

We know that velocity ratio of worm and worm wheel,

$$\text{V.R.} = \frac{DT}{2r} = \frac{300 \times 40}{2 \times 50} = 120$$

and $$\text{M.A.} = \frac{W}{P} = \frac{1800}{24} = 75$$

$\therefore$ Efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{75}{120} = 0.625 = 62.5\% \qquad \textbf{Ans.}$$

Example 11.10. *In a double threaded worm and worm wheel, the number of teeth on the worm wheel is 60. The diameter of the effort wheel is 250 mm and that of the load drum is 100 mm. Calculate the velocity ratio. If the efficiency of the machine is 50%, determine the effort required to lift a load of 300 N.*

Solution. Given : No. of threads (n) = 2; No. of teeth on the worm wheel (T) = 60; Diameter of effort wheel = 250 mm; Diameter of load drum = 100 mm or radius (r) = 50 mm; Efficiency (η) = 50% = 0.5 and load to be lifted (W) = 300 N.

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Velocity ratio of the machine

We know that velocity ratio of a worm and worm wheel,

$$\text{V.R.} = \frac{DT}{2nr} = \frac{250 \times 60}{2 \times 2 \times 50} = 75 \qquad \textbf{Ans.}$$

Effort required to lift the load

Let P = Effort required to lift the load.

We also know that mechanical advantage,

$$\text{M.A.} = \frac{W}{P} = \frac{300}{P}$$

and efficiency,

$$0.5 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{\frac{300}{P}}{75} = \frac{4}{P}$$

or

$$P = \frac{4}{0.5} = 8 \text{ N} \qquad \textbf{Ans.}$$

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11.15. SIMPLE SCREW JACK

It consists of a screw, fitted in a nut, which forms the body of the jack. The principle, on which a screw jack works, is similar to that of an inclined plane.

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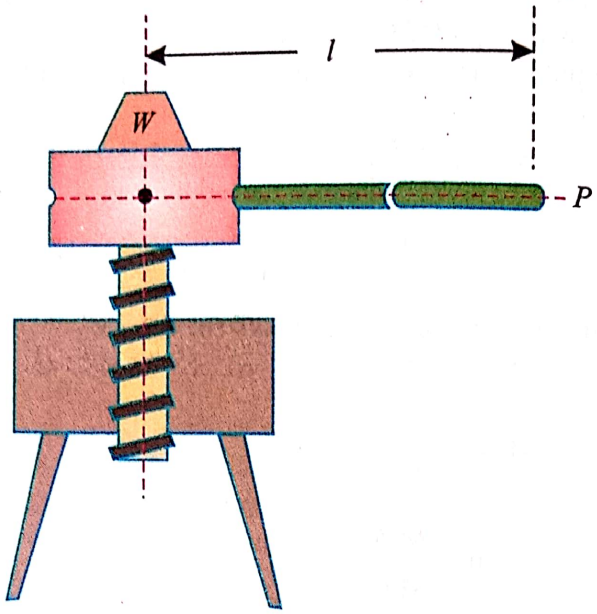


Fig. 11.14. Simple screw jack.

Fig. 11.14. shows a simple screw jack, which is rotated by the application of an effort at the end of the lever, for lifting the load. Now consider a single threaded simple screw jack.

Let l = Length of the effort arm,

p = Pitch of the screw,

W = Load lifted, and

P = Effort applied to lift the load at the end of teh lever.

We know that distance moved by the effort in one revolution of screw,

$= 2\pi l$

and distance moved by the load $= p$ ...(i)

$$\therefore \quad \text{Velocity ratio} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}} = \frac{2\pi l}{p} \quad \text{...(ii)}$$

Now $$\text{M.A.} = \frac{W}{P} \quad \text{...(iii)}$$

and efficeincy, $$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$ ...as usual

...as usual

Note: The value of P *i.e.*, the effort applied may also found out by the relation :

$$*P = W \tan (\alpha + \phi)$$

where W = Load lifted

$$\tan \alpha = \frac{p}{\pi d}$$

and $\tan \phi = \mu$ = Coefficient of friction.

Example 11.20. *A screw jack has a thread of 10 mm pitch. What effort applied at the end of a handle 400 mm long will be required to lift a load of 2 kN, if the efficiency at this load is 45%.*

Solution. Given: Pitch of thread (p) = 10 mm; Length of the handle (l) = 400 mm; Load lifted (W) = 2 kN = 2000 N and efficiency (n) = 45% = 0.45.

Let P = Effort required to lift the load.

* For details, please refer to Art. 8.19.

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We know that velocity ratio

$$\text{V.R.} = \frac{2\pi l}{p} = \frac{2\pi \times 400}{10} = 251.3$$

and

$$\text{M.A.} = \frac{W}{P} = \frac{2000}{P}$$

We also know that efficiency,

$$0.45 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{\frac{2000}{P}}{251.3} = \frac{7.96}{P}$$

$$P = \frac{7.96}{0.45} = 17.7 \text{ N} \quad \textbf{Ans.}$$

11.16. DIFFERENTIAL SCREW JACK

It is an improved form of a simple screw Jack in which the velocity ratio is intensified with the help of a differential screw. In Fig. 11.15. is shown a jack, with a differential screw. The principle on which this machine works, is the same as that of any other differential machine, *i.e.*, action of one part of the machine is subtracted from the action of another part.

In this machine, the differential screw is in two parts, *A* and *B*. Part *A* is threaded both on inside and outside ; whereas the part *B* is threaded on the outside only. The external threads of *A* gear with the threads of the nut *C*, which forms the body of the differential screw jack. The internal threads of *A* gear with the external threads of the screw *B*. Thus the part *A* behaves as a screw for the nut *C* and as a nut for the screw *B*.

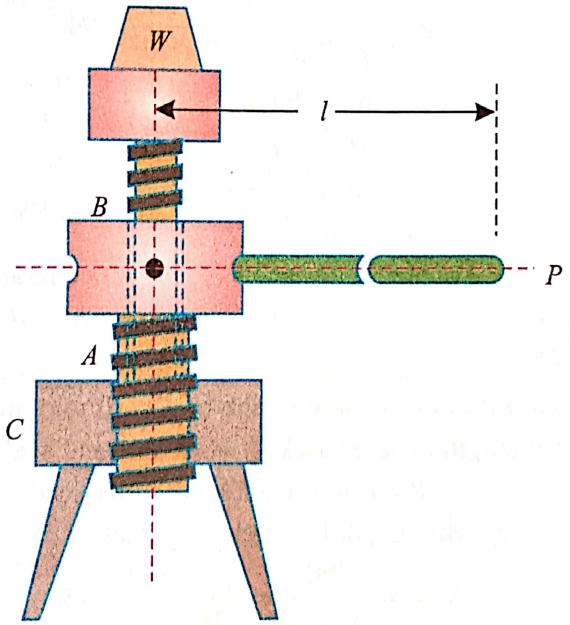


Fig. 11.15. Differential screw jack.

The screw *B* does not rotate, but moves in vertical direction only, and carries the load. When the effort is applied at the lever, the screw *A* rises up and simultaneously the screw *B* goes down. Thus the lift of the load is algebraic sum of the motions of the screw *A* and screw *B*.

Let

p_1 = Pitch of screw *A*,

p_2 = Pitch of the screw *B*,

l = Length of the lever arm,

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W = Load lifted, and

P = Effort applied, at the end of the lever to lift the load.

We know that distance moved by the effort in one revoluton of the lever arm,

$$= 2\pi l \quad \text{...(i)}$$

∴ Upward distance moved by the screw A

$$= p_1$$

and downward distance moved by the screw B

$$= p_2$$

∴ Distance through which the load is lifted

$$= p_1 - p_2 \quad \text{...(ii)}$$

∴ $$\text{Velocity ratio} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}} = \frac{2\pi l}{p_1 - p_2} \quad \text{...(iii)}$$

Now $$\text{M.A.} = \frac{W}{P} \quad \text{...as usual}$$

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} \quad \text{...as usual}$$

Example 11.21. *A differential screw jack has pitch of 12 mm and 10 mm and 30 mm arm length. What will be the efficiency of the machine, if it can lift a load of 7.5 kN by an effort of 30 N.*

Solution. Given: Pitch of the screw (p_1) = 12 mm and (p_2) = 10 mm ; Arm length of screw jack (l) = 300 mm ; Load lifted (W) = 7.5 kN = 7500 N and effort (P) = 30 N.

We know that velocity ratio

$$\text{V.R.} = \frac{2\pi l}{p_1 - p_2} = \frac{2\pi \times 300}{12 - 10} = 942$$

and $$\text{M.A.} = \frac{W}{P} = \frac{7500}{30} = 250$$

∴ Efficiency $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{250}{942} = 0.265 = 26.5\%$$ **Ans.**

Example 11.22. *In a differential screw jack, the screw threads have pitch of 10 mm and 7 mm. If the efficiency of the machine is 28%, find the effort required at the end of an arm 360 mm long to lift a load of 5 kN.*

Solution. Given: Pitch of the screw jack (p_1) = 10 mm and (p_2) = 7 mm ; Efficiency (η) = 28% = 0.28; Arm length of screw jack (l) = 360 mm and load lifted (W) = 5 kN = 5000 N.

Let P = Effort required to lift the load.

We know that velocity ratio of a differential screw Jack.

$$\text{V.R.} = \frac{2\pi l}{p_1 - p_2} = \frac{2\pi \times 360}{10 - 7} = 754$$

and $$\text{M.A.} = \frac{W}{P} = \frac{5000}{P}$$

We also know that efficiency,

$$0.28 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{\frac{5000}{P}}{754} = \frac{6.63}{P}$$

or $$P = \frac{6.63}{0.28} = 23.7 \text{ N}$$ **Ans.**

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or

11.9. SINGLE PURCHASE CRAB WINCH

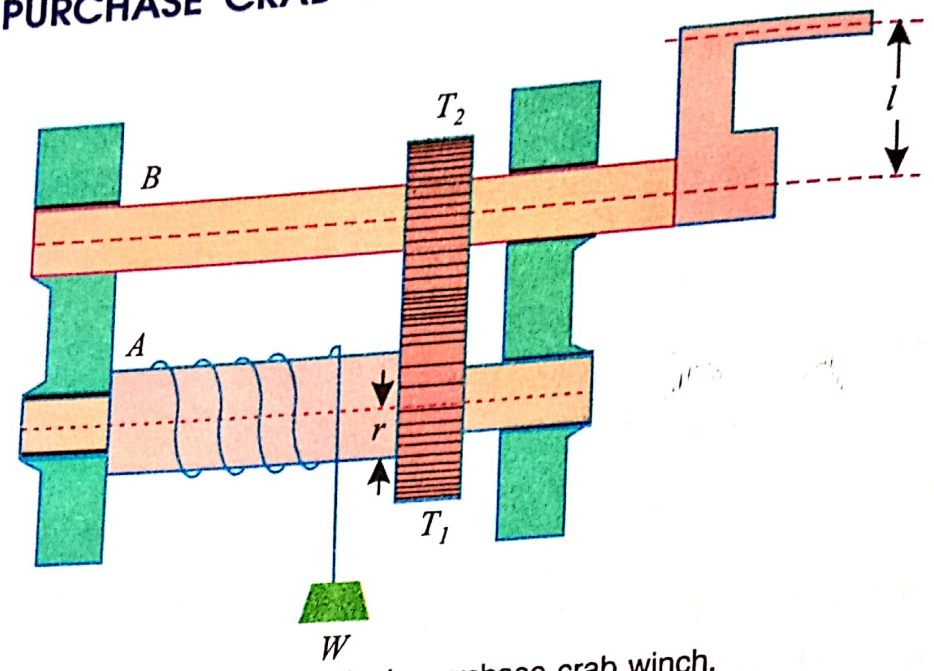


Fig. 11.7. Single purchase crab winch.

In single purchase crab winch, a rope is fixed to the drum and is wound a few turns round it. The free end of the rope carries the load W. A toothed wheel A is rigidly mounted on the load drum. Another toothed wheel B, called pinion, is geared with the toothed wheel A as shown in Fig. 11.7.

The effort is applied at the end of the handle to rotate it.

Let T_1 = No. of teeth on the main gear (or spur wheel) A,

T_2 = No. of teeth on the pinion B,

l = Length of the handle,

r = Radius of the load drum.

W = Load lifted, and

P = Effort applied to lift the load.

We know that distance moved by the effort in one revolution of the handle,

$$= 2\pi l \quad \ldots(i)$$

$\therefore$ No. of revolutions made by the pinion B

$$= 1$$

and no. of revolutions made by the wheel A

$$= \frac{T_2}{T_1}$$

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∴ No. of revolutions made by the load drum

$$= \frac{T_2}{T_1}$$

and distance moved by the load $= 2\pi r \times \frac{T_2}{T_1}$ ...(*ii*)

$$\therefore \quad \text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}} = \frac{2\pi l}{2\pi r \times \frac{T_2}{T_1}} = \frac{l}{r} \times \frac{T_1}{T_2} \quad ...(iii)$$

Now $\text{M.A.} = \frac{W}{P}$ ...as usual

and efficiency, $\eta = \frac{\text{M.A.}}{\text{V.R.}}$ ...as usual

Example 11.13. *In a single purchase crab winch, the number of teeth on pinion is 25 and that on the spur wheel 100. Radii of the drum and handle are 50 mm and 300 mm respectively. Find the efficiency of the machine and the effect of friction, if an effort of 20 N can lift a load of 300 N.*

Solution. Given: No. of teeth on pinion (T_2) = 25; No. of teeth on the spur wheel (T_1) = 100; Radius of daum (r) = 50 mm; Radius of the handle or length of the handle (l) = 300 mm; Effort (P) = 20 N and load lifted (W) = 300 N.

Efficiency of the machine

We know that velocity ratio

$$\text{V.R.} = \frac{1}{r} \times \frac{T_1}{T_2} = \frac{300}{50} \times \frac{100}{25} = 24$$

and $$\text{M.A.} = \frac{W}{P} = \frac{300}{20} = 15$$

∴ Efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{15}{24} = 0.625 = 62.5\%$$ **Ans.**

Effect of friction

We know that effect of friction in terms of load,

$$F_{(\text{load})} = (P \times \text{V.R.}) - W = (20 \times 24) - 300 = 180 \text{ N}$$

and effect of friction in terms of effort,

$$F_{(\text{effort})} = P - \frac{W}{\text{V.R.}} = 20 - \frac{300}{24} = 7.5 \text{ N}$$

It means that if the machine would have been ideal (*i.e.* without friction) then it could lift an extra load of 180 N with the same effort of 20 N. Or it could have required 7.5 N less force to lift the same load of 300 N. **Ans.**

Example 11.14. *A single purchase crab winch, has the following details:*

Length of lever = 700 mm
Number of pinion teeth = 12
Number of spur gear teeth = 96
Diameter of load axle = 200 mm

It is observed that an effort of 60 N can lift a load of 1800 N and an effort of 120 N can lift a load of 3960 N.

What is the law of the machine ? Also find efficiency of the machine in both the cases.

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Solution. Given: Length of lever (l) = 700 *mm*; No. of pinion teeth (T_2) = 12; No. of spur geer teeth (T_1) = 96 and dia of load axle = 200 mm or radius (r) = 200/2 = 100 mm.

(*i*) *Law of the machine*

When $P_1 = 60$ N, $W_1 = 1800$ N and when $P_2 = 120$ N, $W_2 = 3960$ N.

Substituting the values of P and W in the law of the machine *i.e.*, $P = mW + C$

$$60 = (m \times 1800) + C \quad \text{...(i)}$$

and

$$120 = (m \times 3960) + C \quad \text{...(ii)}$$

Subtracting equation (*i*) from equation (*ii*)

$$60 = m \times 2160$$

or

$$m = \frac{60}{2160} = \frac{1}{36}$$

Now substituting this value of m in equation (*i*),

$$60 = \left(\frac{1}{36} \times 1800\right) + C = 50 + C$$

$\therefore$

$$C = 60 - 50 = 10$$

and now substituting the value of $m = 1/36$ and $C = 10$ in the law of machine,

$$P = \frac{1}{36}W + 10 \qquad \textbf{Ans.}$$

(*ii*) *Efficiencies of the machine in both the cases*

We know that velocity ratio

$$\text{V.R.} = \frac{l}{r} \times \frac{T_1}{T_2} = \frac{700}{100} \times \frac{96}{12} = 56$$

and mechanical advantage in the first case

$$\text{M.A.} = \frac{W_1}{P_1} = \frac{1800}{60} = 30$$

$\therefore$ Efficiency

$$\eta_1 = \frac{\text{M.A}}{\text{V.R.}} = \frac{30}{56} = 0.536 = 53.6\% \qquad \textbf{Ans.}$$

Similarly, mechanical advangate in the second case,

$$\text{M.A.} = \frac{W_2}{P_2} = \frac{3960}{120} = 33$$

$\therefore$ Efficiency

$$\eta_2 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{33}{56} = 0.589 = 58.9\% \qquad \textbf{Ans.}$$

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11.10. DOUBLE PURCHASE CRAB WINCH

A double purchase crab winch is an improved form of a single purchase crab winch, in which the velocity ratio is intensified with the help of one more spur wheel and a pinion. In a double purchase crab winch, there are two spur wheels of teeth T_1 and T_2 and T_3 as well as two pinions of teeth T_2 and T_4.

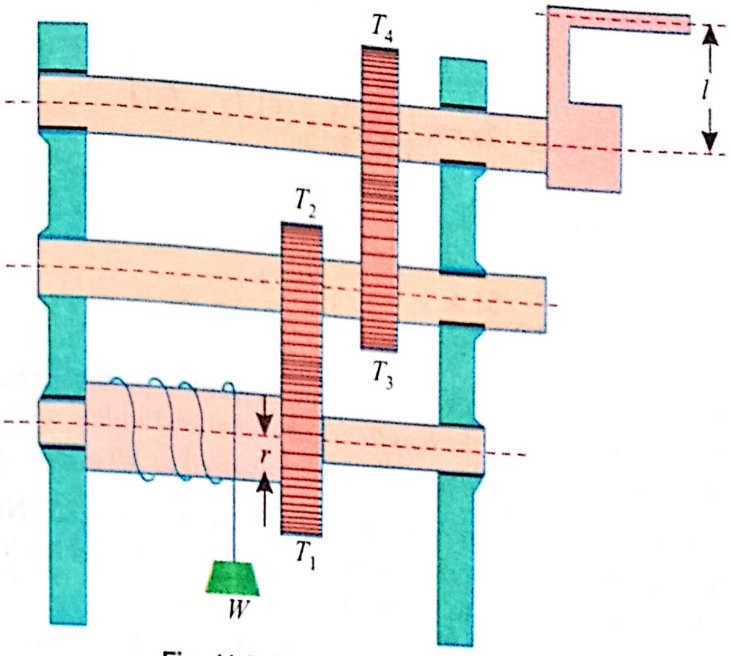


Fig. 11.8. Double purchase crab winch.

The arrangement of spur wheels and pinions are such that the spur wheel with T_1 gears witht he pinion of teeth T_2. Similarly, the spur wheel with teeth T_3 gears with the pinion of the teeth T_4. The effort is applied to a handle as shown in Fig. 11.8.

Let T_1 and T_3 = No. of teeth of spur wheels,

T_2 and T_4 = No. of teeth of the pinions

l = Length of the handle,

r = Radius of the load drum,

W = Load lifted, and

P = Effort applied to lift the load, at the end of the handle.

We know that distance moved by the effort in one revolution of the handle,

$$= 2\pi l \qquad ...(i)$$

∴ No. of revolutions made by the pinion 4

$$= 1$$

and no. of revolutions made by the wheel 3

$$= \frac{T_4}{T_3}$$

∴ No. of revolutions made by the pinion 2

$$= \frac{T_4}{T_3}$$

and no. of revolutions made by the wheel 1

$$= \frac{T_2}{T_1} \times \frac{T_4}{T_3}$$

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∴ Distance moved by the load

$$= 2\pi r \times \frac{T_2}{T_1} \times \frac{T_4}{T_3}$$

$$\therefore \quad \text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}}$$

$$= \frac{2\pi l}{2\pi r \times \frac{T_2}{T_1} \times \frac{T_4}{T_3}} = \frac{l}{r}\left(\frac{T_1}{T_2} \times \frac{T_3}{T_4}\right)$$

Now $$\text{M.A.} = \frac{W}{P}$$

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$

Example 11.15. *In a double purchase crab winch, teeth of pinions are 20 and 25 and of spur wheels are 50 and 60. Length of the handle is 0.5 metre and radius of the load drum is 0. metre. If efficiency of the machine is 60%, find the effort required to lift a load of 720 N.*

Solution. Given: No. of teeth of pinion $(T_2) = 20$ and $(T_4) = 25$; No. of teeth of s wheel $(T_1) = 501$ and $(T_3) = 60$; Length of the handle $(l) = 0.5$ m; Radius of the load drum $(r) = 0.$ m; Efficiency $(\eta) = 60\% = 0.6$ and load to be lifted $(W) = 720$ N.

Let P = Effort required in newton to lift the load.

We know that velocity ratio

$$\text{V.R.} = \frac{1}{r}\left(\frac{T_1}{T_2} \times \frac{T_3}{T_4}\right) = \frac{0.5}{0.25}\left(\frac{50}{20} \times \frac{60}{25}\right) = 12$$

and $$\text{M.A.} = \frac{W}{P} = \frac{720}{P}$$

∴ Efficiency $$0.6 = \frac{\text{M.A.}}{\text{V.R.}} = \frac{\frac{720}{P}}{12} = \frac{60}{P}$$

or $$P = \frac{60}{0.6} = 100 \text{ N}$$ **Ans.**

Example 11.16. *A double purchase crab used in a laboratory has the following dimensio*

Diameter of load drum = *160 mm*

Length of handle = *360 mm*

No. of teeth on pinions = *20* and *30*

No. of teeth on spur wheels = *75* and *90*

When tested, it was found that an effort of 90 N was required to lift a load of 1800 N and effort of 135 N was required to lift a load of 3150 N. Determine :

(a) *Law of the machine,*

(b) *Probable effort to lift a load of 4500 N,*

(c) *Efficiency of the machine in the above case,*

(d) *Maximum efficiency of the machine.*

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Solution. Given: Dia of load drum = 160 mm or radius (r) = 100/2 = 80 mm; Length of handle (l) = 360 mm; No. of teeth on pinions (T_2) = 20 and (T_4) = 30 and no. of teeth on spus wheels (T_1) = 75 and (T_3) = 90.

When $P = 90$ N, $W = 1800$ N when $P = 135$ N, $W = 3150$ N

(a) Law of the machine,

Substituting the values of P and W in the law of the machine, *i.e.*, $P = m\,W + C$

$$90 = (m \times 1800) + C \quad ...(i)$$

$$135 = (m \times 3150) + C \quad ...(ii)$$

and Subtracting equation (i) from equation (ii),

$$45 = m \times 1350$$

or $$m = \frac{45}{1350} = \frac{1}{30}$$

Now substituting this value of m in equation (i),

$$90 = \frac{1}{30} \times 1800 + C = 60 + C$$

$\therefore$ $$C = 90 - 60 = 30$$

and now substituting the value fo m and C in the law of the machine,

$$P = \frac{1}{30} W + 30 \qquad \textbf{Ans.}$$

(b) Effort to lift a load of 4500 *N*

Substituting the value of W equal to 4500 N in the law of the machine,

$$P = \left(\frac{1}{30} \times 4500\right) + 30 = 180 \text{ N} \qquad \textbf{Ans.}$$

(c) Efficiency of the machine in the above case

We know that velocity ratio

$$\text{V.R.} = \frac{l}{r}\left(\frac{T_1 \times T_3}{T_2 \times T_4}\right) = \frac{360}{80}\left(\frac{75}{20} \times \frac{90}{30}\right) = 50.6$$

and $$\text{M.A.} = \frac{W}{P} = \frac{4500}{180} = 25$$

$\therefore$ Efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{25}{50.6} = 0.494 = 49.4\% \qquad \textbf{Ans.}$$

(d) Maximum efficiency of the machine

We also know that maximum efficiency of the machine,

$$\eta_{max} = \frac{1}{m \times \text{V.R.}} = \frac{1}{\frac{1}{30} \times 50.6} = 0.593 = 59.3\% \qquad \textbf{Ans.}$$

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or

11.5. WESTON'S DIFFERENTIAL PULLEY BLOCK

It consists of two blocks A and B. The upper block A has two pulley (P_1 and P_2) one having its diameter a little larger than that of the other. The Pulleys turn together as one pulley, *i.e.*, both of them behave as one pulley with two grooves. The lower block B also carries a pulley, to which the load W is attached.

An endless (*i.e.*, a continuous) chain passes round the pulleys then round the lower block pulley and finally round the pulley P_2 (*i.e.*, smaller of the upper pulleys). The remaining chain hangs slack and is joined to the first portion of the chain as shown in Fig. 11.3.

The effort P is applied to the chain passing over the pulley P_1 (*i.e.*, larger of upper pulleys) as shown in Fig. 11.3. To prevent the chain from slipping, projection are provided in the grooves of both the upper pulleys.

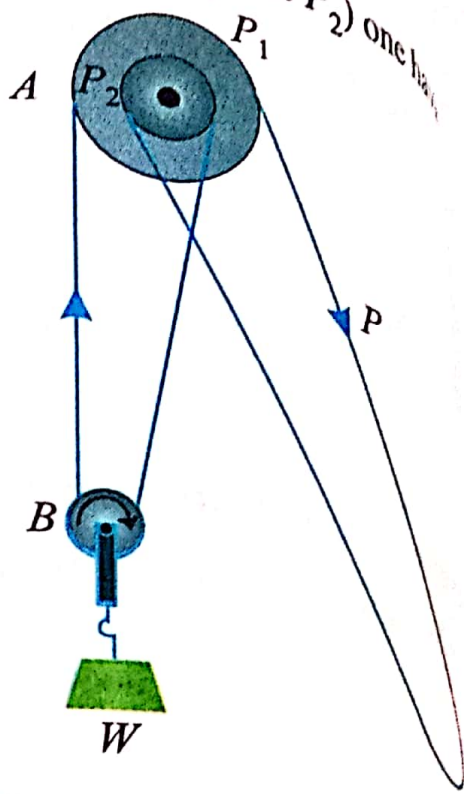


Fig. 11.3. Differential pulley b

Let

D = Diameter of the pulley P_1,

d = Diameter of the pulley P_2,

W = Weight lifted, and

P = Effort applied to lift the weight.

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We know that displacement of the effort in one revolution of the upper pulley block,

$$= \pi D \qquad \ldots(i)$$

This is also equal to the length of the chain pulled over the larger pulley. Since the smaller pulley also turns with the larger one, therefore length of the chain released by the smaller pulley

$$= \pi d$$

$\therefore$ Net shortening of the chain $= \pi D - \pi d = \pi (D - d)$

This shortening of chain will be equally divided between the two portions of the chain, supporting the load. Therefore distance through which the load will move up

$$= \frac{1}{2} \times \pi (D - d) = \frac{\pi}{2}(D - d) \qquad \ldots(ii)$$

$\therefore$ $$\text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}}$$

$$= \frac{\pi D}{\frac{\pi}{2}(D - d)} = \frac{2D}{D - d} \qquad \ldots (iii)$$

Now $$\text{M.A.} = \frac{W}{P} \qquad \ldots \text{as usual}$$

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} \qquad \ldots\text{as usual}$$

Notes: 1. Sometimes, the size of the upper pulleys are given in terms of radius (instead of diameters).

If R = Radius of the larger pulley, and

r = Radius of the smaller pulley.

Then $$\text{V.R.} = \frac{2R}{R - r}$$

2. Sometimes, the diameters of upper pulleys are not given and the relatives sizes are expressed in terms of number of teeth.

If T_1 = No. of teeth of the larger pulley, and

T_2 = No. of teeth of the smaller pulley

Then $$\text{V.R.} = \frac{2T_1}{T_1 - T_2}$$

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11.11. SIMPLE PULLEY

A simple pulley is a wheel of metal or wood, with a groove around its circumference, to receive rope or chain. The pulley rotates freely about its axle, which passes through its centre and is perpendicular to its surface plane. This axle is supported by a metal or a wooden frame, called block as show in Fig. 11.9. Following assumptions are made in the study of pulley system, which are quite reasonable from the practical point of view :

1. The weight of the pulley block is small as compared to the weight to be lifted, and thus may be neglected in calculations.
2. The friction between the pulley surface and the string is negligible, and thus the tension in the two sides of the rope, passing round the pulley, may be taken to be equal.

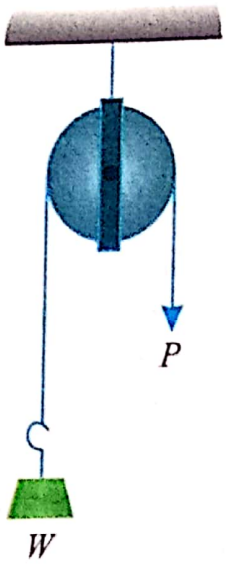


Fig. 11.9. Simple pulley.

A little consideration will show, that in a simple pulley, its mechanical advantage as well as velocity ratio is 1 under the assumed conditions mentioned above. The only advantage of a simple pulley is that the effort can be applied Fig. 11.10 (*a*), (*b*) and (*c*). Simple pulleys are generally used in certain mechanical advantage and efficiency.

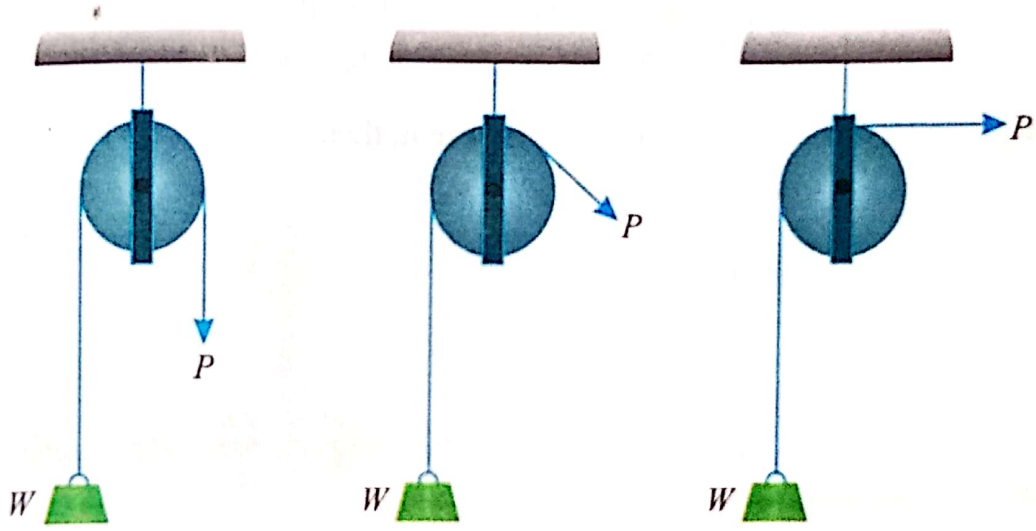


Fig. 11.10. Force applied in different directions.

Though there are many typed of pulleys used by engineers, yet the following system of pulleys are commonly used :

1. First system of pulleys.
2. Second system of pulleys.
3. Third system of pulleys.

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11.12. FIRST SYSTEM OF PULLEYS

In Fig. 11.11 is shown the first system of pulleys. In this system, the pulleys are so arranged that there are as many strings as there are pulleys. The end of each string is fastened to a rigid ceiling; while the other end passing round the bottom periphery of the pulley, is fastened to the next higher pulley.

The load is attached to the bottom-most pulley ; whereas the effort is applied to the far end of the string passing round the last pulley. Another pulley (no. 5) is used just to change the direction of the effort.

The velocity ratio of the system may be obtained by considering a unit motion of the load. In this case, let the weight W be raised by x metres. Since the loads is supported on both sides of the string, thus this slackness of x metres will have to be taken up by the pulley 2. If the relative position of the pulley 2, with respect to the pulley 1, is to remain undisturbed, then the pulley 2 should move upwards through a distance of $2x$ metres.

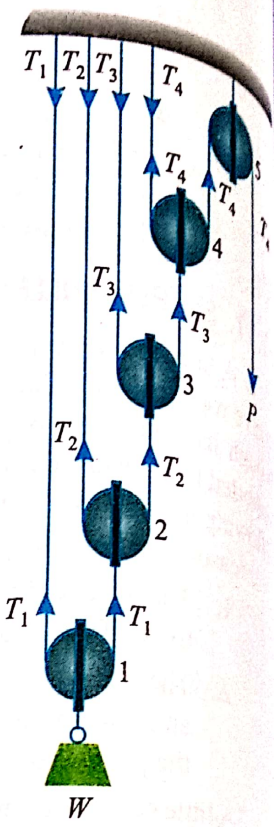


Fig. 11.11. First system of pulleys

Now this upward movement of pulley 2 through a distance of $2x$ metres will cause a total slackness of $2 \times (2x) = 2^2x$ metres in the string, which has to be taken up by the pulley 3. Thus the pulley 3 should move upwards through a distance of 2^2x metres, thus causing a slackness of $2 \times (2^2x) = 2^3x$ in the string passing round the pulley 4. Thus the pulley 4 should move upwards through a distance of 2^3x metres causing a slackness of $2 \times (2^3x) = 2^4x$ meters, which must be taken up by the free end of the string to which the effort is applied. Thus the effort must mov through a distanece of 2^4x metres.

$$\therefore \quad \text{V.R.} = \frac{\text{Distance moved by the effort}}{\text{Distance moved by the load}}$$

Thus, in general, is there are n pulley in the system, then

$$\text{V.R.} = 2^n$$

Now $$\text{M.A.} = \frac{W}{P}$$ ...as usua

and efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$ ...as usua

Example. 11.17. *In a system of pulleys of the first type, there are three pulleys, and a weigh of 320 N can be lifted by an effort of 50 N.*

Find the efficiency of the machine and the amount of friction.

Solution. Given : No. of pulleys (n) = 3 ; Weight lifted (W) = 320 N and effort (P) = 50 N

Efficeincy of the machine

We know that velocity ratio of first system of pulleys.

$$\text{V.R.} = 2^n = 2^3 = 8$$

and $$\text{M.A.} = \frac{W}{P} = \frac{320}{50} = 6.4$$

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∴ Efficiency, $$\eta = \frac{\text{M.A.}}{\text{V.R.}} = \frac{6.4}{8} = 0.80 = 80\%$$ **Ans.**

Amount of friction

We know that amount of friction in terms of load,

$$F_{(\text{load})} = (P \times \text{V.R.}) - W = (50 \times 8) - 320 = 80 \text{ N}$$ **Ans.**

and amount of friction in terms of effort,

$$F_{(\text{effort})} = P - \frac{W}{\text{V.R.}} = 50 - \frac{320}{8} = 10 \text{ N}$$ **Ans.**

SECOND SYSTEM OF PULLEYS

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